Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Drawing Parallel Lines

Two lines are parallel when they run side by side and never meet, no matter how far they are extended. You can tell they are parallel because they keep the same distance apart all along their length. In a diagram, parallel lines are often marked with matching little arrows.

In this section you will learn how to draw a line through a given point that is parallel to a given line, using only a ruler, a set square, and a pencil. The set square is the triangle-shaped tool with a right angle (a 90° corner) at one end. It does the important job for us: because its straight edge stays fixed against the ruler, we can slide it along without turning it, and the lines it draws stay parallel.

Here is the idea behind the method. When you draw any straight line and then draw another line that cuts across it (called a transversal), the two lines are parallel exactly when the angles they make with that crossing line are equal. In the construction below we use this fact in a simple mechanical way: sliding a set square along a ruler keeps its edge at the same angle the whole time, so the line you draw is always at the same angle to the ruler — and that guarantees the new line is parallel to the first.

Worked example: Construct a line through the point P that is parallel to the line AB.

Step 1 — Draw the given line AB and mark the point P anywhere off it. Keep P a sensible distance away, not too close and not too far.

Step 2 — Place your set square so that one of its straight edges lies exactly on top of the line AB.

Step 3 — Keeping the set square pressed down firmly, place your ruler against the other straight edge of the set square so the ruler touches it along its whole length. Hold the ruler very still.

Step 4 — While holding the ruler in place, slide the set square along the ruler towards P. Do not let it twist or lift: it must stay pressed against the ruler the whole time.

Step 5 — When the top edge of the set square reaches P, keep it there and draw a straight line through P along that edge. This new line is parallel to AB.

Step 6 — Mark the new line with a little arrow (the same style of arrow as on AB) to show the two lines are parallel.

Because you slid the set square without turning it, the edge you drew along stayed at exactly the same angle to the ruler as the line AB was. That equal angle is what guarantees the two lines are parallel.

Let us look at the reasoning once more, because it is the whole secret of the construction.

Imagine a line CD drawn across the line AB (so CD cuts AB at some point). Now imagine the same line CD drawn across your new line. If the angle that CD makes with AB is exactly the same as the angle that CD makes with the new line, then the two lines are parallel. This is what we depend on.

In the set-square method, the ruler plays the part of CD: it is the fixed reference line that both AB and your new line cross. When you slide the set square along the ruler without turning it, you are drawing a line that meets the ruler at exactly the same angle that AB meets the ruler. Same crossing line, same angle — so the two lines must be parallel.

A useful check when you finish: parallel lines are the same distance apart everywhere. Use your ruler to measure the gap between AB and the new line at two different places. If both measurements are equal, your construction has worked.

Practice

1. Draw a straight horizontal line and label it AB. Choose a point P above the line, about halfway along it. Using only a ruler and a set square, construct a line through P that is parallel to AB. Explain in one sentence why the method works.

2. Line XY is drawn slanting downwards. Construct a line through a point Q, placed below and to the right of the line, that is parallel to XY. What single thing must you avoid doing while sliding the set square, and why does that matter?
3. Two students each draw a line through point P. Student A slides the set square along the ruler without turning it. Student B turns the set square a little as it slides. Only one of the two lines is guaranteed parallel to the original line. Which student, and why?

Section 2: Drawing a Perpendicular Line with Ruler and Compass

A perpendicular line is a line that meets another line at a right angle — an angle of exactly 90°. Think of the corner of a sheet of paper, or the way the two arms of a plus sign (+) cross each other. When two lines are perpendicular, we can mark the right angle with a small square drawn at the corner.

In construction work you often need to draw a perpendicular line. You will learn two situations:
1. Dropping a perpendicular from a point that lies above a line, down to meet the line.
2. Erecting (raising) a perpendicular at a point that lies on the line itself.

Both methods use only a ruler and a compass, and both rely on the same simple idea: the point you want lies exactly on the bisector of a segment you have drawn. The two tools you need are your ruler (for drawing straight lines) and your compass (for drawing circles and arcs). Keep the compass opening fixed whenever you want two arcs that are exactly the same size.

Before you start, set out your tools. Place the ruler on your paper, and with a sharp pencil draw a long straight line across it. Label this line with two letters, for example AB. Mark a point P somewhere above the line. You will construct the perpendicular from P down to the line, meeting it at a new point you will call M.

The steps below build a perpendicular step by step. Follow them slowly with your own line and point, and check each step against your drawing as you go.

Worked Example 1 — Dropping a perpendicular from a point above a line

We construct the perpendicular from point P down to the line AB.

Step 1: With the point of your compass on P, open the compass a little wider than the distance down to the line. Draw an arc that cuts the line AB in two places. Call these two cut points X and Y. (Choosing a compass opening bigger than the drop makes sure the arc actually reaches the line.)

Step 2: Put the compass point on X. Open the compass to about two-thirds of the distance between X and Y, and draw an arc below the line, roughly beneath the middle of XY.

Step 3: Keeping the compass at exactly the same opening, put the point on Y and draw another arc below the line. This second arc crosses the first. Call the crossing point Q.

Step 4: Use your ruler to draw the straight line from P to Q. Where this line crosses AB, call the point M.

The line PM is perpendicular to AB. Check it: PM meets AB at right angles, and the small corner between them is a perfect square. The point M is exactly beneath P, and PM stands upright on AB.

Worked Example 2 — Erecting a perpendicular at a point on the line

Sometimes the point where the perpendicular must stand is already on the line. Suppose point O lies on the line AB, and we want to raise a perpendicular straight up from O.

Step 1: Put the compass point on O. With any convenient opening, draw arcs that cut AB on both sides of O. Call these two cuts X (on the left) and Y (on the right). The opening can be anything, but X and Y must be equal distances from O.

Step 2: Open the compass wider. Put the point on X and draw an arc above the line.

Step 3: Keeping the same wider opening, put the point on Y and draw another arc above the line. The two arcs cross. Call the crossing point R.

Step 4: Draw the straight line from O through R with your ruler. Extend it well above the line.

The line OR is perpendicular to AB, standing straight up from O. Again, the corner at O is a right angle, marked by a small square.

Practice

1. Draw a line AB on your paper. Mark a point P above it. Write down the first two steps you would take to drop a perpendicular from P down to meet AB.
2. In Worked Example 2, when you erect a perpendicular at point O on line AB, why must the two cuts X and Y be equal distances from O?
3. After you have drawn the arc from X below the line in a perpendicular construction, what must you do with your compass before drawing the arc from Y, and why?
4. Explain in one sentence what it means for a constructed line to be perpendicular to another, and name the two tools you used to build it.

Section 3: Cutting a Line Segment into Two Equal Halves

To **bisect** a line segment means to cut it into two exactly equal parts. The point where it is cut is called the **midpoint**, and it lies exactly halfway between the two ends. We also use the word "bisect" for angles (cutting an angle into two equal angles), but here we focus on the line segment itself.

You will need a ruler, a sharp pencil, and a pair of compasses. Here is the way to bisect a line segment, step by step.

Step 1 — Draw the segment. Use your ruler to draw a straight line segment AB. We want to find the point exactly in the middle of A and B.

Step 2 — Open the compasses. Set the compasses to a width a little more than half the length of AB. How do you know it is more than half? Guess roughly where the middle is, and open the compasses wider than that guess. (If you use a width less than half, the two arcs you are about to draw will not meet.)

Step 3 — Draw the first arc. Place the compass point on A. Draw an arc that swings above and below the segment. Keep the compasses at the same setting — do not change the width.

Step 4 — Draw the second arc. Without changing the compass width, place the point on B and draw another arc above and below. The two arcs cross at two points: one above the segment and one below it. Mark these two crossing points.

Step 5 — Join the crossings. Use your ruler to draw a straight line through the two crossing points. This line is called the **perpendicular bisector**: it crosses AB at a right angle, and the point where it meets AB is the midpoint. That midpoint bisects (cuts into two equal halves) the segment.

The line you drew through the two arc crossings is exactly halfway along AB, so the two smaller segments on either side of the midpoint are equal in length. Try measuring them with your ruler — they will match.

Worked example — Bisecting a 6 cm segment

Let us bisect a line segment AB that measures 6 cm.

1. Draw AB with your ruler so that it is 6 cm long. Label the left end A and the right end B.
2. Set your compasses to a width of about 4 cm. This is more than half of 6 cm, which is 3 cm, so the arcs will meet. (Anything wider than 3 cm works.)
3. Place the compass point on A and draw a wide arc that curves above and below AB.
4. Without changing the width, place the point on B and draw another arc. The two arcs cross at two points — one above the segment and one below it. Call the top crossing P and the bottom crossing Q.
5. Draw a straight line through P and Q. Where this line meets AB is the midpoint, M.

Now check: AM should measure 3 cm and MB should measure 3 cm. The line PQ has cut the 6 cm segment into two equal 3 cm pieces, so AB is bisected. The line PQ is also perpendicular to AB, which is why it is called the perpendicular bisector — but our goal here, the midpoint M, is what bisects the segment.

Practice

1. A line segment CD is 10 cm long. You follow the bisection steps: draw an arc from C, draw a matching arc from D, join the two crossing points, and the joining line meets CD at point M. How long is CM, and how long is MD?

2. While bisecting segment PQ, you set your compasses so that the width is 1 cm, but PQ is 5 cm long. What goes wrong when you try to draw the arcs, and why?
3. A perpendicular bisector is drawn on a line segment XY, and it meets XY at the midpoint N. If XN measures 7 cm, what is the full length of XY?

Section 4: Building a Right Angle: Constructing 90°

A right angle is an angle of exactly 90°. It is the angle you see at the corner of a sheet of paper or a building. When two lines meet at 90°, we call them perpendicular. In construction work, we use a ruler and a pair of compasses to draw a 90° angle without needing a protractor, and it comes out very accurately every time.

Here is the idea behind the construction. We start with a straight line and a point on it. From that point we build a right angle using the same compass-and-ruler moves that bisect a straight angle. If you follow the steps in order, you will always finish with two lines that meet at exactly 90°.

What you need:
• A sharp pencil
• A ruler (with straight edges)
• A pair of compasses

The construction has five steps. Take your time with each one and keep the compass opening the same where the steps tell you to.

Step 1 – Draw a horizontal line across your page with the ruler. Mark a point on it and call it A. This is where the corner of your 90° angle will sit.

Step 2 – Put the compass point on A and open it to any convenient width. Draw an arc that crosses the line on both sides of A. Call the two points where the arc meets the line B (left of A) and C (right of A).

Step 3 – Without changing the compass opening, put the compass point on B and draw an arc above the line, roughly in the middle between B and C.

Step 4 – Keeping the compass opening the same, put the compass point on C and draw another arc above the line. This arc must cross the one you just drew. Call the point where the two arcs meet D.

Step 5 – Use your ruler to draw a straight line from A through D. The angle between this new line and your original line at A is exactly 90°.

Why does this work? Because B and C are equal distances from A, and the two arcs above the line are drawn with the same compass opening, so B and C are equally far from D as well. That makes the line AD perfectly upright, meeting the horizontal line at a right angle.

Worked Example

Construct a 90° angle at a point P on a horizontal line.

Step 1 – Draw a horizontal line and mark point P on it.
Step 2 – With the compass point on P, draw an arc that cuts the line on both sides. Call the cuts Q (left) and R (right).
Step 3 – Without changing the compass opening, put the point on Q and draw an arc above the line.
Step 4 – With the same opening, put the point on R and draw an arc above the line so it crosses the first arc. Call the crossing point S.
Step 5 – Join P to S with your ruler.

The line PS makes an angle of 90° with the horizontal line. The angle between your horizontal line and PS at point P is the right angle you wanted.

Practice

1. On a fresh part of your page, draw a horizontal line and mark a point T in the middle of it. Using only your ruler and compasses, construct a 90° angle at T. Describe each step you took, and name the points you used, so that someone reading your steps could build the same angle.

2. Look at the construction you have been doing. In Step 3 you drew an arc from one side of A, and in Step 4 you drew an arc from the other side of A with the same compass opening. Why must the compass opening stay exactly the same between these two steps?

Section 5: Constructing a 60° Angle

A 60° angle is one of the special angles you can build with just a compass and a straightedge (ruler). The trick is to use the fact that a circle is made up of 360°, and each of the six equal "wedges" around a point spans 60°.

Here is the method, step by step.

Step 1 — Draw a ray.
On your paper, draw a ray starting at a point A and passing through a point B. This gives you a straight line to work from. The point A is where your 60° angle will sit (the vertex).

Step 2 — Make the first arc.
Put the metal point of the compass exactly on A. Open the compass to any convenient width, and draw a short arc that crosses the ray AB. Call the point where the arc cuts the ray P.

Step 3 — Make the second arc.
Do not change the compass opening at all — keep it exactly the same width. Now move the compass point onto P and draw another arc. It will cross your first arc. Call this crossing point Q.

Step 4 — Join A to Q.
Draw a straight ray from A through Q. The angle between the ray AB and the ray AQ is 60°.

Why does this work? Because in Steps 2 and 3 you kept the compass opening the same, the distance AP equals PQ equals AQ. So triangle APQ has all three sides equal — it is an equilateral triangle. Every angle in an equilateral triangle is 60°, so the angle at A is 60°.

The two most common slips are changing the compass width between Steps 2 and 3 (everything falls apart), and drawing the second arc on the wrong side of the ray. Make sure both arcs are drawn with the same opening, and it works first time.

Worked example: Construct a 60° angle at the point A on a line.

Setup: Draw a ray AB on your paper.

1. Place the compass point on A and draw an arc crossing AB. Mark this crossing point P.
2. Keep the compass exactly as it is. Place the point on P and draw a second arc, so it crosses the first arc. Mark the crossing Q.
3. Draw the ray from A through Q.

The angle BAQ is your 60° angle. If you have a protractor handy, you can check it — it should read 60°.

Notice what you must never do between steps 2 and 3: you must not widen or narrow the compass. Keeping the opening fixed is what makes AP = PQ = AQ, which is what forces the angle to be exactly 60°.

Practice

1. Draw a ray starting at point A. Using only a compass and a ruler, construct a 60° angle at A. In your own words, write down the four steps you took.
2. A friend tries to construct a 60° angle but after drawing the first arc they open the compass wider before drawing the second arc. Explain, in your own words, why this will not give a 60° angle.
3. After constructing a 60° angle at A between ray AB and ray AQ, how many times does the angle 60° fit into a full turn of 360° around point A? Use this to explain why six copies of your angle fill a full circle.

Check your work

Attempt every question before you read this page.

Drawing Parallel Lines

1. You slide the set square along the ruler without turning it, so the new line meets the ruler at the same angle as AB. Because the new line and AB make equal angles with the same crossing line (the ruler), the two lines are parallel. Check by measuring the gap in two places — the distances should be equal.

   Place the set square so one straight edge lies on AB. Rest the ruler against the other edge and hold it still. Slide the set square along the ruler up to P without twisting it, then draw a straight line through P along the set square's edge. The line through P is parallel to AB.
2. You must avoid letting the set square twist or lift away from the ruler while you slide it. If it twists, its edge no longer keeps the same angle to the ruler, so the line you draw would meet the ruler at a different angle and would not be parallel to XY.

   The method works for a slanting line exactly as for a horizontal one: put the set square's edge on XY, rest the ruler against the other edge and hold it firmly, slide the set square down to Q without turning it, and draw the line through Q. The equal angles with the ruler guarantee parallelism.
3. Student A's line is guaranteed parallel, because keeping the set square at the same angle to the ruler makes the new line meet the ruler at the same angle as the original line. Student B turned the set square, which changes that angle, so the line is not guaranteed to be parallel.

   Equal angles with the same crossing line (the ruler) give parallel lines. Student A preserves the angle by sliding without turning; Student B changes the angle by turning, so the equal-angle condition no longer holds.

Drawing a Perpendicular Line with Ruler and Compass

1. Step 1: Put the compass point on P, open it a little wider than the distance from P down to the line, and draw an arc that cuts AB in two places, X and Y. Step 2: Put the compass point on X and draw an arc below the line.

   The perpendicular is built with two equal arcs. First, from P draw one arc that reaches the line and cuts it twice — those two cuts give you X and Y. Then you begin the equal-arc step by drawing an arc from X below the line.
2. Because they are drawn with the same compass opening from O, so XO and OY are equal. That equality is what places the perpendicular exactly above the middle, standing straight up from O.

   Both arcs in Step 1 are made with the same compass opening centred on O, so the distances OX and OY are the same. A perpendicular erected at O must come up from the midpoint of the segment between the two arc cuts, and equal cuts give exactly that midpoint.
3. Keep the compass at exactly the same opening as the arc from X. Then the arc from Y will cross the first one, and the two arcs are the same size so the crossing point sits on the perpendicular.

   The two lower arcs must be identical in size so that they are symmetric about the perpendicular. If you changed the opening, the crossing point would drift off the line that should be perpendicular. Keeping the opening fixed guarantees the arcs meet at a point directly below the starting point.
4. A perpendicular line meets the other line at a right angle of 90°, and it is built using only a ruler and a compass.

Cutting a Line Segment into Two Equal Halves

1. CM = 5 cm and MD = 5 cm. Bisecting cuts the segment into two equal halves, so each half is $10 \div 2 = 5$ cm.

   The midpoint M splits CD into two equal parts. So $CM = 10 \div 2 = 5$ cm and $MD = 10 \div 2 = 5$ cm. The two halves must be equal, which is the whole point of bisecting.
2. The arcs do not meet, so you cannot find the crossing points. The compass width must be more than half the length of the segment (more than 2.5 cm for a 5 cm segment) or the two arcs drawn from P and Q will not cross.

   Half of 5 cm is 2.5 cm. With the compasses set to only 1 cm, the arc drawn from P stays within 1 cm of P, and the arc drawn from Q stays within 1 cm of Q. There is a gap in the middle between 1 cm and 4 cm along the segment, so the arcs never touch. The compass width always has to be more than half the segment length for the two arcs to meet.

3. XY = 14 cm. Since N is the midpoint, XN and NY are equal, so NY = 7 cm, and the whole segment is 7 + 7 = 14 cm.

   The perpendicular bisector meets the segment at its midpoint, which cuts it into two equal halves. So NY = XN = 7 cm. Adding the two equal halves gives XY = 7 + 7 = 14 cm.

Building a Right Angle: Constructing 90°

1. Construct the 90° angle at T exactly as in the worked example: (1) draw an arc centred at T cutting the line at two points; (2) keeping the same opening, draw arcs from each of those two points above the line so they cross; (3) join T to the crossing point. The line you draw meets the original horizontal line at 90°.

   Step 1 – Draw the horizontal line and mark T.
   Step 2 – Put the compass point on T and draw an arc that cuts the line on both sides. Call the cuts U (left) and V (right).
   Step 3 – Without changing the opening, put the point on U and draw an arc above the line.
   Step 4 – With the same opening, put the point on V and draw an arc above the line so it crosses the first one. Call the crossing point W.
   Step 5 – Join T to W with the ruler. The angle between the horizontal line and TW is exactly 90°.
2. Because the two points on the line (like Q and R) must be equally far from the crossing point above the line. If the opening changed, the crossing point would be closer to one side, and the line you finally draw would lean over and not be at 90°.

   In Step 3, the arc centred at Q marks every point that is a fixed distance from Q. In Step 4, the arc centred at R marks every point the same distance from R, but only if the compass opening has not changed. Their crossing point D is therefore the same distance from Q and from R. Since Q and R are themselves equally far from A, the line AD stands exactly upright and meets the horizontal line at 90°. Changing the opening would break the equal distances and tilt the final line.

Constructing a 60° Angle

1. a. Draw the ray AB.
   b. With the compass point on A, draw an arc crossing AB; mark the crossing P.
   c. Keeping the same compass opening, put the point on P and draw an arc crossing the first; mark the crossing Q.
   d. Draw the ray from A through Q. The angle BAQ is 60°.

   The key point is that the compass opening stays the same the whole way, so AP = PQ = AQ. That makes triangle APQ equilateral, so its angle at A is 60°.
2. If the compass width is changed, then the distances AP, PQ and AQ are no longer all equal, so triangle APQ is no longer equilateral and its angle at A is no longer 60°.

   The 60° angle comes from building an equilateral triangle, and an equilateral triangle needs all three sides equal. Changing the compass opening breaks that equality.
3. 360° ÷ 60° = 6, so a 60° angle fits exactly six times around a point, filling a full turn.

   A full turn is 360°. Since each angle is 60°, six of them give 6 × 60° = 360°, which is exactly one full turn.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-0e20fe3d	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____